MATLAB-Based Numerical Simulation of BCR-ABL1 Expression Dynamics and Targeted Therapy Response in Chronic Myeloid Leukemia

Course: Numerical Analysis / MATLAB Project

1. Abstract

Chronic myeloid leukemia (CML) is a type of myeloproliferative cancer caused predominantly by the activation of BCR-ABL1 fusion oncogene as a consequence of the Philadelphia chromosome translocation. Such an activation leads to increased survival and proliferation of the involved myeloid cells through a constitutively active tyrosine kinase. Since chronic myeloid leukemia is characterized by its distinct molecular cause, the use of BCR::ABL1 tyrosine kinase inhibitors (TKIs), such as imatinib, dasatinib, nilotinib, bosutinib, asciminib and ponatinib, makes it a suitable subject for numeric modeling [1-3]. This work involves development of a simplified ODE-based numerical model of dynamics of BCR-ABL1 mRNA expression levels and the protein/kinase activity. Three scenarios are implemented: for normal hematological cells, untreated chronic myeloid leukemia cells, and treated chronic myeloid leukemia cells. The choice of scenarios is based on the availability of the public gene expression dataset (GSE33075) with normal donor samples and CML patients prior to imatinib treatment and post-treatment [4]. Approximation is done using Euler's method; MATLAB ode45 function is utilized for comparison purposes as a more advanced solver. The simulations are expected to show higher BCR-ABL1 activity in untreated CML and lower activity after treatment.

2. Introduction / Biological Background

Chronic myeloid leukemia (CML) is a hematologic cancer that affects all blood cells which develop from myeloid precursors. The disease exists because myeloid white blood cells acquire the ability to grow beyond normal limits and continue to exist. The principal genetic feature of CML involves the Philadelphia chromosome which results from a chromosome 9 and 22 reciprocal translocation. This process creates the BCR-ABL1 fusion gene which functions as the primary genetic cause of the disorder.

The BCR-ABL1 fusion gene produces a defective tyrosine kinase protein. CML cells contain active BCR-ABL1 kinase but healthy cells maintain control over their tyrosine kinase signaling pathways. This constant signaling activates cellular pathways which enable leukemic cells to multiply and survive while resisting typical growth regulation. BCR-ABL1 activity plays a major role in CML progression and treatment response.

CML targeted therapies primarily use BCR::ABL1 TKIs as their main treatment option. The National Cancer Institute lists TKIs such as imatinib mesylate, dasatinib, nilotinib, bosutinib and asciminib as CML treatments, and recent CML management reviews emphasize that BCR::ABL1 TKIs remain central to modern CML therapy [1,2]. The 2025 European

LeukemiaNet recommendations also highlight treatment response monitoring, resistance, and BCR::ABL1 mutation testing as important parts of CML management [3].

The GEO dataset GSE33075 was used to support the biological conditions used in the simulations. The dataset title states that "Imatinib therapy of chronic myeloid leukemia restores the expression levels of key genes for DNA damage and cell cycle progression." The study includes human expression data which demonstrates three different groups: normal donors and Ph-positive CML patients who have not received imatinib treatment and CML patients who have undergone one month of therapy [4]. Although this project does not perform

3. Aim of the Project

The main goal of this project is to build a simple numerical model for BCR-ABL1 mRNA and protein dynamics in CML. Three different scenarios will be simulated, including the normal hematological cell with low BCR-ABL1 activity, the untreated CML with high BCR-ABL1 activity and the treated CML where the therapy lowers the activity of BCR-ABL1.

The particular goals of this project are to:

- provide the explanation of the biological function of BCR-ABL1 in CML;
- validate the biological data by using the set GSE33075 and applying it to the normal, untreated and treated CML scenarios;
- develop the ordinary differential equations-based model of the dynamics of BCR-ABL1 mRNA/protein;
- identify parameter values used in the three scenarios under consideration;
- apply the Euler approximation for finding numerical solutions step by step;
- compare the solution obtained with MATLAB ode45; and
- interpret the numerical dynamics in light of CML development and treatment.

Thus, the expected results will demonstrate that the untreated CML scenario will have the largest activity value, and the normal cell – the smallest. The third one should have lower activity since inhibition corresponds to TKIs.

4. Mathematical Model

The model used to describe BCR-ABL1 gene expression kinetics is a two variable ODE model comprising the levels of BCR-ABL1 mRNA and protein/activities. The simplicity of the model is attributed to the goal of demonstrating numerical modelling and not the reproduction of CML biology as seen in patients.

Symbol	Meaning	Biological interpretation

M(t)	BCR-ABL1 mRNA level	Amount of BCR-ABL1 transcript at time t
P(t)	BCR-ABL1 protein/activity level	BCR-ABL1 tyrosine kinase protein/activity at time t
Parameter	**Meaning**	**Biological interpretation**
k_m	mRNA production rate	Transcription of BCR-ABL1 mRNA
d_m	mRNA degradation rate	Natural breakdown of BCR-ABL1 mRNA
k_p	protein/activity production rate	Production of BCR-ABL1 protein/activity from mRNA
d_p	protein/activity degradation rate	Natural degradation/inactivation of protein/activity
u	treatment inhibition rate	Reduction of BCR-ABL1 activity due to therapy

The mRNA equation is:

$$dM/dt = k_m - d_m M$$

This equation means that mRNA increases through transcription and decreases through degradation. The protein/activity equation is:

$$dP/dt = k_p M - d_p P - uP$$

In the equation above, k_p M is the rate of protein/activity synthesis through mRNA production, d_p P is the rate of natural protein degeneration/inactivation, and uP stands for the inhibition caused by medication. For untreated individuals, the value of u is zero, but for treated individuals, $u>0$. This model considers the following factors: steady production of mRNA, linear decay, proportionate production of protein to the amount of mRNA, and one factor of medication inhibition.

5. Parameters and Simulation Scenarios

M(0) = 0.1 and P(0) = 0.1 were used to denote the minimal basal molecular activity during the initial time frame. The time range was between 0 and 50 arbitrary time units, and the value of h for the Euler method was 0.1. These values of parameters are assumed for the sake of comparison only.

Scenario	k_m	d_m	k_p	d_p	u	Expected behavior
Normal hematopoietic cell	0.5	0.3	0.6	0.4	0	Low mRNA and protein/activity
Untreated CML cell	2.0	0.2	1.5	0.2	0	Highest mRNA and protein/activity
Treated CML cell	2.0	0.2	1.5	0.2	0.8	Reduced protein/activity due to therapy

In the normal condition case, the rate of production of proteins/activities is kept low to signify low levels of BCR-ABL1 activities. In the untreated CML condition, the production rate of mRNAs and proteins/activities is kept high to simulate an oncogenic situation. Finally, in the treated CML condition, the production rate is kept similar to that of the CML condition, while a parameter value of u=0.8 is added to indicate TKI-type inhibition.

6. Numerical Methods

The model is solved using numerical techniques since the objective of the project is to create a realistic simulation of BCR-ABL1 mRNA and its protein activity/levels over time based on varying biological circumstances. The Euler technique is applied to demonstrate how the simple approximation process works, whereas the ode45 function of MATLAB is utilized as a more accurate solver. These methods help connect the mathematical model to the biological system.

6.1 Euler's Method

Euler's method is simple and useful for demonstrating numerical approximation, although it is less accurate than ode45. For a general ODE $dy/dt = f(t,y)$, with $y(t_0) = y_0$, Euler's method estimates the next value using:

$$y_{n+1} = y_n + h\, f(t_n, y_n)$$

where h is the step size, y_n is the current approximation, and $f(t_n, y_n)$ is the current slope. Educational numerical-method sources describe Euler's method as the simplest method for approximating ODE solutions by following the current slope over a step of length h [6].

Applied to the BCR-ABL1 mRNA equation, Euler's method gives:

$$M_{n+1} = M_n + h(k_m - d_m M_n)$$

Applied to the protein/activity equation, Euler's method gives:

$$P_{n+1} = P_n + h(k_p M_n - d_p P_n - u P_n)$$

The Euler algorithm used is: define M(0), P(0), the time interval, and h; select the parameter set for normal, untreated CML, or treated CML; calculate the mRNA slope $k_m - d_m M_n$; update M; calculate the protein/activity slope $k_p M_n - d_p P_n - u P_n$; update P; repeat until t = 50; then plot M(t) and P(t). A smaller h generally improves accuracy but increases the number of computations. Euler's method is useful for demonstration, but because it is first-order, it is less accurate than higher-order adaptive methods such as ode45.

The MATLAB code should generate the mRNA and protein/activity curves for the three scenarios and compare Euler's method with ode45. The results should then be interpreted biologically rather than treated as exact clinical predictions, because the parameter values are assumed for educational numerical simulation.

6.2 MATLAB ode45 Solver

ode45 is a built-in function in MATLAB for numerically solving Ordinary Differential Equation systems; it is based on the Runge-Kutta integration methods. The model used in this project, the BCR-ABL1 gene expression model, was simulated using this built-in numerical solver under normal, untreated CML and treated CML conditions. Unlike Euler's method, ode45 automatically adjusts the step size, which gives smoother and more stable numerical results.

The model presented in this project represents time changing dynamics of BCR-ABL1 mRNA, and activity levels at various biological states. Therefore, a built-in numerical solver like ode45 can provide a computational tool to simulate these dynamics when the analytic solutions are too complicated to acquire.

Using ode45 solver in MATLAB it was simulated, and the result was compared with Euler's method and the behavior under different treatment inhibition strengths on the activity behavior was observed. A higher activity level is anticipated in untreated CML and a much lower activity level is predicted after the treatment effect, as shown below.

7. MATLAB Implementation

7.1 Numerical Simulation Workflow

A computational environment of MATLAB was used in the numerical simulation of the proposed BCR-ABL1 mathematical model. The simulation workflow included parameter initializations, numerical solution of the ODE systems using Euler's method and ode45, visualization of BCR-ABL1 mRNA and protein/activity in normal state, treated CML without therapeutic treatment and treated CML. Initial conditions, and the biological parameters of transcription, degradation, protein production, and treatment inhibition were defined before numerical simulations.

7.2 MATLAB Code Structure

The MATLAB implementation of this simulation involved parameter initialization, numerical solvers, plotting functions, and comparative simulations using both Euler's method and ode45. Euler's method used update equations with time steps and loop structures, while ode45 utilized MATLAB's built-in adaptive numerical integration. Other simulations of treatment inhibition with different strength values were run as well to study therapy-associated BCR-ABL1 activity dynamics.

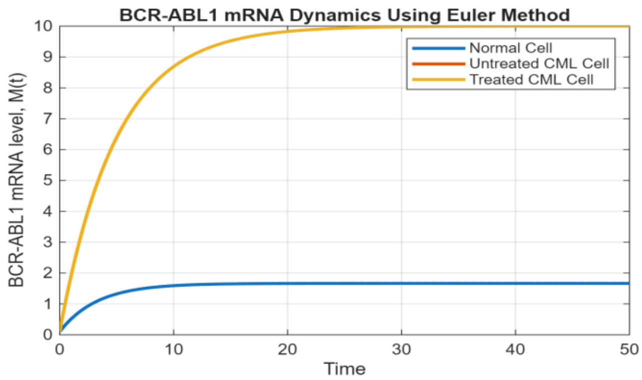
7.3 GEO Transcriptomic Dataset Analysis

A public GEO transcriptomic dataset of CML (GSE33075) was downloaded and used to analyze the effect of treatment on the gene expression in CML samples. It was divided into healthy control, treated CML, and untreated CML. Analysis of treatment-associated down-regulation of genes was then carried out, with averages from different groups computed and expression plots of downregulated probes generated using MATLAB visualization tools.

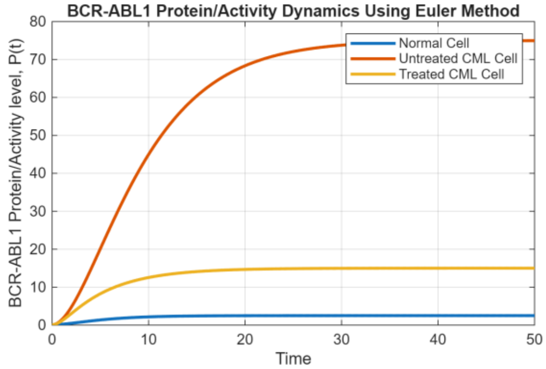
8. Results

8.1 Euler Method Simulation Results

The dynamics of BCR-ABL1 mRNA and protein/activity were simulated using Euler's method for normal, untreated CML, and treated CML conditions.



In figure 1 we see high levels of mRNA under untreated CML conditions. A treatment reduced the simulated activity levels compared with untreated CML.



From figure 2 we can observe that the protein/activity level was the highest under untreated CML, while under treated conditions it was significantly lower.

8.2 MATLAB ode45 Simulation Results

MATLAB ode45 was used to simulate BCR-ABL1 mRNA and protein/activity for normal, untreated CML, and treated CML conditions.

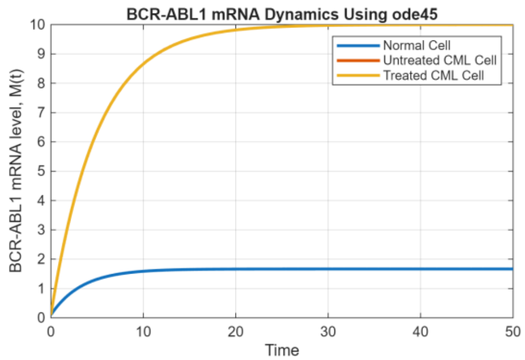
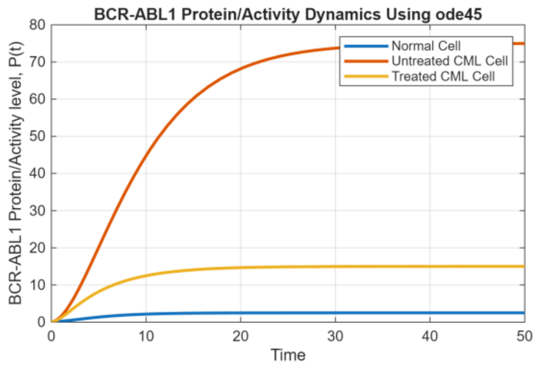


Figure 3 shows the elevated mRNA dynamics under untreated CML and based inhibited activity under treatment-based inhibition.



In figure 4 we can see high protein/activity under untreated CML and very low activity under treated conditions.

8.3 Euler Method and ode45 Comparison

Comparisons were made between Euler's method and ode45 through simulations.

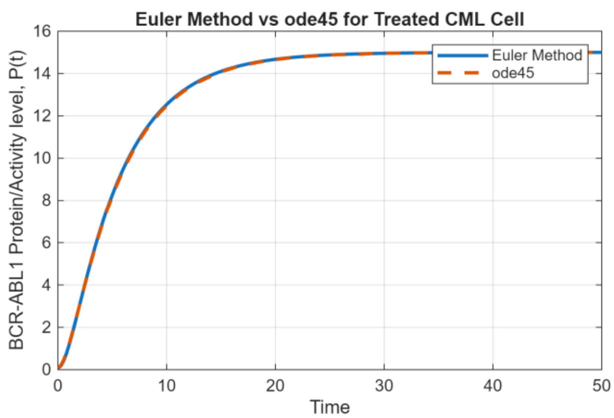


Figure 5 shows that Euler's method and ode45 followed very similar overall trends. Both numerical methods are performing similarly when dealing with the BCR-ABL1 protein/activity dynamics.

8.4 Effect of Treatment Strength

A different approach was taken when analyzing the data; the different inhibition strengths of the treatment were used.

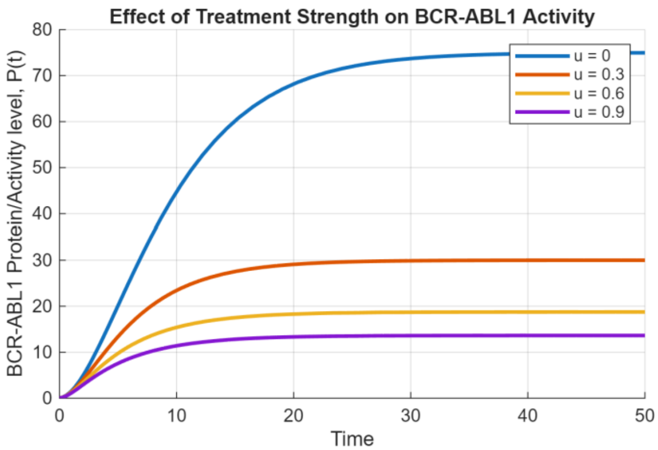


Figure 6: As treatment inhibition increased, the simulated protein/activity levels became lower.

8.5 GEO Transcriptomic Dataset Analysis

With the usage of the GEO transcriptomic dataset, multiple probes have been found to have down regulation after treatment in CML.

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Top Downregulated Genes

"227510_x_at"  Change = -3.401
"208835_s_at"  Change = -2.545
"223578_x_at"  Change = -2.531
"202581_at"  Change = -2.435
"203372_s_at"  Change = -2.322
"203373_at"  Change = -2.245
"219387_at"  Change = -2.243
"229120_s_at"  Change = -2.043
"211734_s_at"  Change = -2.025
"224567_x_at"  Change = -1.980
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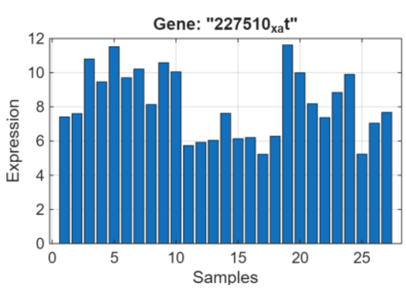

Figure 7 illustrates the expression profile of one of the most downregulated probes identified during the analysis.

9. Discussion

9.1 Numerical Interpretation of Euler and ode45 simulations

Both the numerical solution and the ode45 simulations indicated an increase in activity levels for the suggested model of the BCR-ABL1 model under untreated CML conditions and inhibition of activity under treatment conditions. ode45 produced smoother curves, but Euler's method showed the same general behavior.

9.2 Biological interpretation of treatment dynamics

The decrease in simulated BCR-ABL1 activity under treatment conditions is in line with the activity of tyrosine kinase inhibitors biologically. The decrease in simulated activity levels increased with increasing treatment inhibition strength and demonstrated decreased activity under untreated CML conditions.

9.3 Biological interpretation GEO transcriptomic data

There were several probes that displayed lower gene expression levels after treatment, according to the transcriptomic data analysis performed on the GEO dataset. While the results were primarily qualitative, these gene expression decreases correlated with the expected simulation result.

9.4 Combined Interpretation of all data sets

Overall, combining numerical modeling with transcriptomic analysis helped explain how BCR-ABL1 activity changes in CML. Both Euler's method and MATLAB ode45 determined elevated BCR-ABL1 activity levels during leukemic activity and a decrease in activity upon introduction of treatment-associated inhibition.

10. Limitations

While a biological interpretation of BCR-ABL1 gene expression can be achieved through the mathematical model outlined, the model is an oversimplification of intracellular signal pathways during chronic myeloid leukemia.
Additionally, parameter values for the simulation were chosen on the basis of qualitative biological significance, not clinical parameter modeling. Finally, the GEO transcriptomic study conducted here was intended for qualitative illustration of general expression changes in response to treatment, not for a complete bioinformatics differential-expression analysis pipeline.

11. Conclusion

In this project a computational and mathematical model of BCR-ABL1 gene expression dynamics during chronic myeloid leukemia was simulated using Euler's method in MATLAB's ode45. Simulations exhibited elevated BCR-ABL1 gene expression under leukemic state, and decreased gene expression in the treated state. Treatment-induced expression changes were observed in public GEO transcriptomics data and were found to be in qualitative agreement with simulation results. Overall, the project showed how numerical methods and computational biology can be used to study gene expression behavior in cancer.

References

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